Antony Raj A

AI ML Engineer

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About Me

AI & ML Engineer with 3+ years of experience building scalable applications, machine learning solutions, and AI applications. Experienced in **Agentic AI, LLMs, RAG, LangGraph, and MCP** for cloud operations, incident management, root-cause analysis, and automated remediation. Proficient in **Python, FastAPI, Django, machine learning, deep learning, data engineering, AI,** and multi-cloud technologies.

Professional Experience

AI ML Engineer
Xaasio Pvt Ltd | Coimbatore — July 2023 – Present

- Developed scalable **Python, Django, and FastAPI** applications for enterprise data, cloud, billing, and AI-driven solutions.
- Developed **Machine Learning, Deep Learning, Generative AI, RAG, and Agentic AI** solutions for forecasting, knowledge retrieval, automation, and intelligent operations.
- Built BSS and FinOps solutions for **multi-cloud billing, pricing, cost visibility, resource-level cost analysis, and optimisation** across public cloud and on-premises environments.
- Developed an AI-powered BSS/CMP Assistant using LLMs and agent-based workflows for natural-language cloud and FinOps operations.
- Developed an **OpenStack SRE Agent** using **LangGraph, MCP, LLMs**, and automation to enable node-level incident detection, root cause analysis, remediation, and health verification.
- Designed data processing pipelines, REST APIs, database integrations, and automation workflows to deliver scalable enterprise solutions.

Educational Qualification

MBA , Karpagam College Of Engineering | Coimbatore
7.9 CGPA — Nov 2021 - Apr 2023

BSC(IT), The American college | Madurai
7.8 CGPA — June 2018 - Apr 2021

Skill Set

Platforms	Linux, Windows
Backend	Python
Frameworks & API	FastApi, Django, Flask, Streamlit
Databases	RDBMS (PostgreSQL, MySQL), Hive, Redis

Containers	Docker, Kubernetes
Data Science	Machine Learning, Deep Learning, Gen AI, Agentic AI
Streaming	Kafka & Spark Streaming
Data Processing	Spark and pandas
Data Engineering	Hadoop, Minio S3
CI/CD	Gitlab, Jenkins

Projects

A. Wing AI application

Description

Handled data migration from SAP system to Hive, including data processing, custom logic development for business requirements, data streaming, sales forecasting, tailor-made application building, and data visualization dashboard creation.

Responsibilities

- Managed data migration from SAP to Hive, cleaned datasets, and built custom logic for 214 reports aligned with business requirements.
- Applied machine learning and deep learning techniques for sales forecasting, including time series analysis for trend prediction.
- Developed a tailor-made application using **Django** framework, integrated with LDAP AD for user authentication.
- Built a **RAG** (Retrieval-Augmented Generation) application for user queries, supporting general chatbot functionality and **natural language to SQL** conversion.

B. OpenSRE Agent

Description

Developed an **Agentic AI-powered OpenStack SRE Agent** for node-level incident detection, root-cause analysis (RCA), automated remediation, and infrastructure health verification.

Responsibilities

- Developed an Agentic AI workflow using **Python, LangGraph, LLMs, MCP, Ansible, and OpenStack** APIs for automated incident management.
- Implemented OpenStack node-level incident detection and RCA across controller, compute, and network infrastructure.
- Integrated **Prometheus** metrics, logs, service health, and OpenStack events for incident investigation and root-cause analysis.

- Developed an **OpenStack MCP** Server to expose infrastructure monitoring, diagnostic, and remediation operations as AI tools.
- Integrated **Ansible** automation to execute corrective actions and remediate identified infrastructure issues.
- Implemented the complete Detect → Investigate → RCA → Remediate → Execute → Verify workflow with guardrails and post-remediation health checks.

C. BSS Application – FinOps & AI Assistant

Description

Developed a multi-cloud Business Support System (**BSS**) integrated with CMP for cloud billing, pricing, FinOps cost management, and AI-powered cloud operations across public cloud and on-premises environments.

Responsibilities

- Developed BSS and FinOps APIs using Python, FastAPI, SQLAlchemy, and PostgreSQL for billing, pricing, resource management, and cost analysis.
- Built multi-cloud cost visibility and aggregation across AWS, Azure, GCP, Oracle Cloud, and on-premises infrastructure.
- Implemented cost analysis by organization, project, provider, region, resource type, and service with daily, weekly, and monthly trends.
- Implemented Valkey caching, rate limiting, RBAC, and authorization to improve API performance and secure FinOps functionality.
- Developed an AI-powered BSS/CMP Assistant using LLM and agent workflows for natural-language interaction with cloud, billing, and FinOps operations.